Dominic Le

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GitHub, Webpage

Research Interests

Causation, counterfactuals, cognitive development, computational cognitive science, learning, memory, philosophy of language, philosophy of cognitive science.

Education

M.A. Developmental Psychology and Education. 2026 - 2027
University of Toronto

H.B.Sc. Psychology and Cognitive Science, *High Distinction.* 2022 - 2026
University of Toronto

Ongoing Research

Backwards Counterfactual Reasoning in Children and Adults Sep 2024 –
Supervisor: Ioana Grosu & Patricia Ganea, University of Toronto.
Department of Applied Psychology and Human Development.

Hierarchical Models of Associative Learning Aug 2024 –
Supervisor: Sara Aronowitz, University of Toronto.
Department of Philosophy.

Modelling Causal Reasoning in a Cognitive Architecture Sep 2024 –
Supervisor: Can Mekik, University of Toronto.
Department of Psychology.

Peer-reviewed Publications

Proceedings

Dominic Le, Can Mekik (2026). *A Spreading Activation Model of Causal Reasoning.* Proceedings of the 23rd International Conference on Cognitive Modelling. Forthcoming.

Ioana Grosu, **Dominic Le**, Patricia Ganea (2025). *Backwards Counterfactuals and the Closest Possible World.* Proceedings of the Annual Meeting of the Cognitive Science Society. Vol. 47. 2025. https://escholarship.org/uc/item/0q93p7sc

Conferences & Workshops

Peer-reviewed

Dominic Le, Can Mekik (2026). *A Spreading Activation Model of Causal Reasoning.* Talk to be presented at the International Computational Cognitive Modelling Conference (ICCM, July), Montreal, CA. Poster presented at the Canadian Society for Brain, Behaviour and Cognitive Science Annual Meeting (CSBBCS, June), North York, CA.

Dominic Le, Ioana Grosu, & Patricia Ganea (2026, April). *Children's Backward Counterfactual Logic.* Poster presented at the Cognitive Development Society Bi-Ennial Conference (CDS), Montreal, CA.

Dominic Le (2025). *Causal Stacks: A Structural Framework for Recurrent and Hierarchical Counterfactual Reasoning.* Presented as poster for the 5th European Experimental Philosophy Conference (EXPhi, July), Norwich, UK; the Society for Philosophy and Psychology (SPP, June) Annual Meeting; and the Annual Meeting of the Cognitive Science Society (CogSci, August).

Ioana Grosu, **Dominic Le**, & Patricia Ganea (2025, July). *Backwards Counterfactuals and the Closest Possible World.* Talk presented at CogSci 2025, San Francisco, US.

Ioana Grosu, **Dominic Le**, & Patricia Ganea (2025, June). *Changes to Models of Reality in Backwards Counterfactual Reasoning.* Poster presented at the 51st Annual Meeting of the Society for Philosophy and Psychology (SPP), Ithaca, US.

Non-peer-reviewed

Dominic Le (2026, October). *Backward counterfactual reasoning may not follow the similarity principle.* Talk presented at the Cognitive Science of Language group (CSoL) at the University of Toronto, Toronto, CA.

Dominic Le, Can Mekik (2026, April). *How Search Strategies Explain Directional Asymmetries in Causal Reasoning.* Talk presented at the Psychology Undergraduate Research Community (PURC) Student Symposium at the University of Toronto, Toronto, CA.

Dominic Le, Ioana Grosu, & Patricia Ganea (2025, February). *Time's Asymmetry in Counterfactuals.* Talk presented at the Cognitive Science and Artificial Intelligence Student Association (CASA) Undergraduate Conference at the University of Toronto, Toronto, CA.

Dominic Le (2024, September). *Modeling Children's Development of Modal Thought.* Talk presented at the Collective Undergraduate Research (CURE) Flash Talks at the University of Toronto, Toronto, CA.

Dominic Le, Oksana Tkachman (2024, March). *Spatial Frames of Reference in Signed Languages.* Talk presented at the Western Interdisciplinary Student Symposium on Language Research (WISSLR) at Western University, London, CA.

Software

Quentin Dercon, Rina Harsch, **Dominic Le**, Shannon Yasuda, Cherrie Chang, Melissa Struhl (2025). *snap-sort (Version 1.0.0)* [Software]. In review for jsPsych-contrib. https://github.com/DomDaSquishyBomb/plugin-drag-and-drop

Labwork

Research Assistant, Cognitive Development Jan 2024 –
PI: Ioana Grosu & Patricia Ganea, University of Toronto
Language & Learning Lab

Research Assistant, Psycholinguistics Mar 2023 –Sep 2024
PI: Oksana Tkachman, University of British Columbia
Language & Learning Lab

Research Assistant, Morality and Abnormal Psychology Nov 2023 –Jun 2024
PI: Mark Miller, University of Toronto
Consciousness & Wisdom Studies Lab

Previous Employment

Child's Program Assistant, UofT Family Housing Jun 2024 –Aug 2024

Supported Child Care Worker, BGC Canada Jun 2023 –Sep 2023

Day Camp Coordinator, Cadets Canada Jun 2022 –Aug 2022

Relevant Undergrad Coursework

Programming & Computer Science:	Seminar on Cognitive Architectures: Python
Data, Computation, and the Mind: Python	Intro to A.I.: Python
Software Design: Java	Intro to Computer Programming: Python
Psychology & Cognitive Science:	Minds and Machines
Behavioural Neuroscience	Semantic Theory
Sensation and Perception	Intro to Cognitive Science
Intro to Philosophy: Knowledge and Reality	Intro to Psychology
Intro to Developmental Psychology	Intro to Cognitive Psychology
Statistics & Mathematics:	Treatment of Psychological Data: R
Experimental Design and Theory: R	Math Expres. and Reas. in Computer Science
Psychology Statistics II: JASP	Probability and Statistics II: R
Probability and Statistics I: R	Multivariable Calculus with Proofs
Linear Algebra	Calculus with Proofs

Student Affiliations

Editor, Mentis - UofT Cognitive Science Undergraduate Journal	2024 –
Mentor, CURE - Undergraduate Research Student Group	2024 –
Staff Writer, Demo Magazine - Hart House Music Publication	2023 - 2024
Executive Secretary, CASA - CogSci and A.I. Student Association	2024 –2025
CogSci Course Union Representative, University College Council	2024 –2025

Awards & Scholarships

($7500) University of Toronto Research Excellence Award (UTEA-SSH).	2026
University of Toronto Student Leadership Award.	2026
($7500) University of Toronto Research Excellence Award (UTEA-SSH).	2025
($1500) Victoria College Regents Participation Award.	2025
(Priceless) MAT237 Meme Contest Nominee. x2	2024
($2000) Gyro Club of Nanaimo Jack Ryan Scholarship.	2022
($1250) British Columbia District/Authority Scholarship.	2022
($1250) British Columbia Excellence Scholarship.	2022
($2000) Vancouver Island University Dual Enrollment Scholarship.	2021

Travel Awards

($2250) W. Gordon And Connie Young Marigold Award (CogSci)	2026
($1000) Arriba Travel Award For Cognitive Science. (CogSci)	2026
($500) Dr. Roseann Runte Award. (CDS)	2026
($1000) Susan Macdonald Award. (CogSci)	2025
($250) ASSU Travel Grant. (EXPhi)	2025
($150) SPP Travel Award.	2025

Skills & Characteristics

Natural Languages:	Fluent: English (native), French. Rudimentary: Vietnamese (first-learned).
Computer Languages:	Proficient: R, Python, Java, HTML/CSS Rudimentary: JavaScript, C#, C++.
Certifications:	Standard First Aid - with CPR-C-AED (St. John Ambulance till 2026), Responsible Adult (YMCA, 2023)
Software Tools:	Document-preparation: MS Office, LaTeX (Overleaf/VSCode), Google Workspace. Data-collection: Qualtrics, MTurk, ChildrenHelpingScience, REDCap, jsPsych. Data-processing: RStudio, Pandas+NumPy, JASP. Image-manipulation: Photoshop, Canva Audio-editing: Ableton Live, Logic, Audacity. Video-editing: Premiere, Final Cut, Animate.
Outside Interests:	Songwriting and role-playing video games.